



Micro-Nano Electro Mechanical System

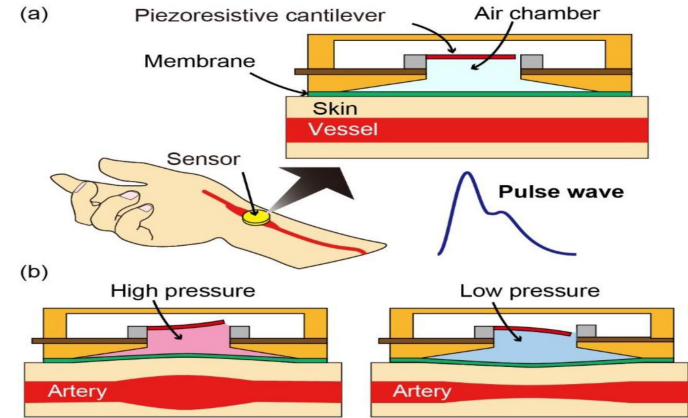
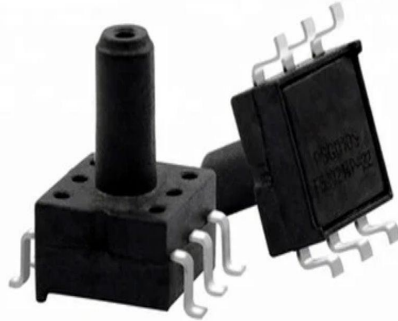
GROUP-6

Presented By

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Introduction to MEMS-NEMS

- M/NEMS stands for **Micro-Nano Electro Mechanical Systems**
- They are very small devices made on **silicon chips**
- They combine **mechanical parts and electronic parts**
- They can sense physical quantities like pressure and motion
- Example: **Cantilever-based pressure sensor**



Cantilever Pressure Sensor

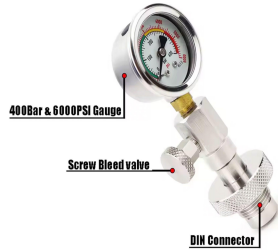
- Uses a micro cantilever beam to sense pressure
- Pressure causes the cantilever to bend
- Bending is converted into an electrical signal

HISTORY OF PRESSURE MEASURING INSTRUMENTS

1. Early and Classical methods



Liquid column manometer

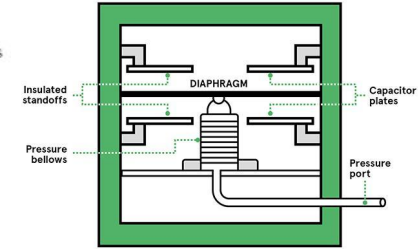
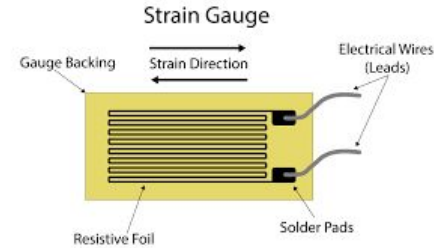


Mechanical elastic gauge

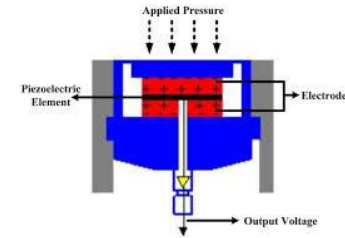


Piston testers

2. Electrical and Electronic pressure sensors



Capacitive Pressure sensor

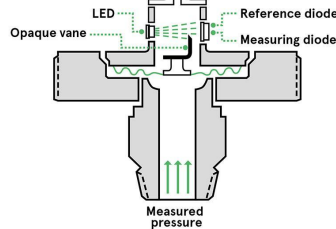


Piezoelectric pressure sensors

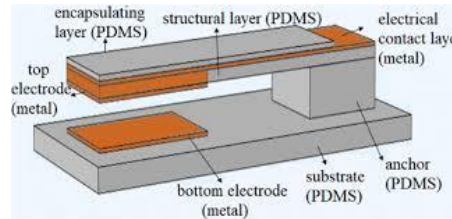
3. Modern MEMS and Integrated Sensors



Silicon piezoresistive



Fiber optic pressure sensors



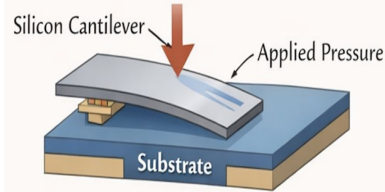
MEMS based Cantilever beam pressure sensor

4. Nanoscale and Quantum Pressure Sensing

- **Quantum defect (NV-center) based sensors:** Nitrogen-vacancy centers in diamond show pressure-dependent shifts in electronic energy levels; optically read using fluorescence and microwaves. Allow **local, sub-micron-scale** sensing of stress/pressure and temperature under extreme conditions, with very high sensitivity.
- **NEMS resonators and nanostructured films**
Nanoscale cantilevers and membranes change resonance frequency or conductance as pressure or surface stress varies. Can be integrated as ultra-small, low-mass pressure or stress sensors with potential for on-chip arrays.

Sensor Type	Typical Pressure Range	Year of Invention
Manometer	10 Pa – 300 kPa	1643 (Torricelli)
Bourdon Tube	100 kPa – 100 MPa	1849 (Eugène Bourdon)
Mechanical Diaphragm	1 kPa – 10 MPa	Early 1900s (~1900–1920)
Electrical Pressure Sensors (piezoresistive, capacitive, piezoelectric)	100 Pa – 50 MPa	1930s–1950s
MEMS Diaphragm Sensors	10 Pa – 1–5 MPa	1980s–1990s
Cantilever-Based MEMS Sensors	1 Pa – 10 kPa	Late 1990s–2000s

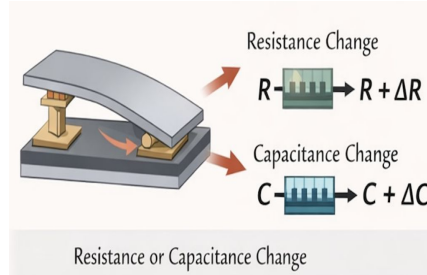
Early Development



Basic Silicon Cantilever Bending from Pressure.

Simple Cantilever Design

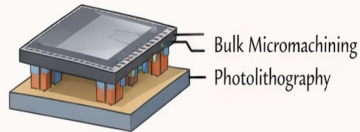
Early sensors used a **single silicon** cantilever beam. When pressure was applied, the beam bent slightly.



Basic Working Principle

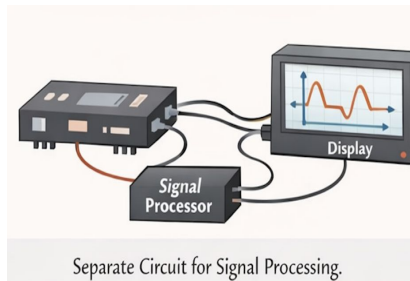
The bending of the cantilever changed **resistance or capacitance**, which was used to measure pressure.

Traditional Fabrication



Basic, Larger Devices.

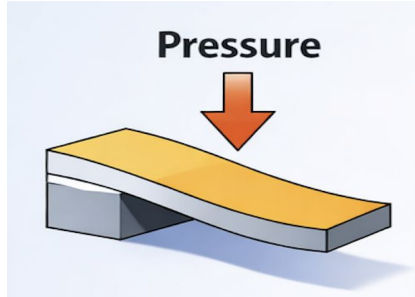
Fabrication was done using basic **micromachining and photolithography**, making the devices bigger in size.



External Electronics Needed

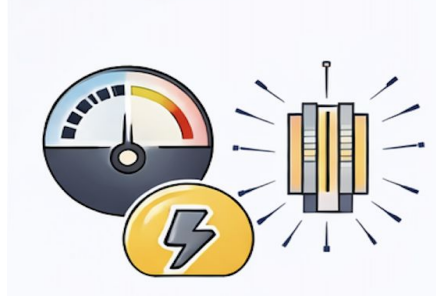
The sensor only detected pressure. Separate external circuits were required to read and process the signal.

Advanced Development



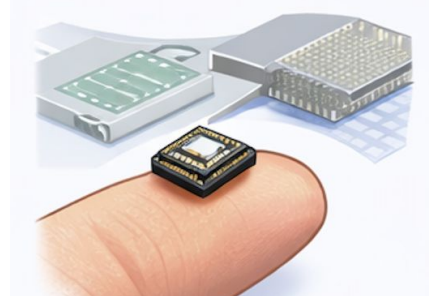
Optimized Cantilever Structure

The cantilever shape and thickness are carefully chosen so it bends in a controlled way under pressure, improving measurement accuracy.



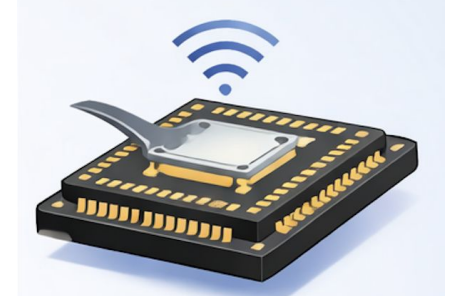
Very High Sensitivity

Advanced sensors can detect extremely small pressure changes using improved capacitive, piezoelectric, or frequency-based methods.



Miniaturized Fabrication

Modern fabrication techniques allow the sensor to be very small, light, and strong, while maintaining consistent performance.



Integrated Electronics

Signal processing circuits are built on the same chip, so the sensor provides a clean and stable output.

State of Availability



Commercial Availability

- MEMS cantilever sensors are available from specialized MEMS manufacturers.
- Applications: biomedical monitoring, microfluidic systems, aerospace instrumentation.



Research Sensors

- Lab-fabricated sensors for high sensitivity and miniaturized applications.
- Fabrication techniques: photolithography, DRIE, surface micromachining.



Key Trends in Development

- Integration with on-chip electronics for signal processing and IoT applications.
- Development of ultra-sensitive, wearable, and portable sensors.



Market Perspective

- Limited commercial variety compared to standard capacitive/piezoresistive MEMS sensors.
- Strong research focus on precision, compactness, and smart sensing capabilities.

Gaps in Conventional MEMS Cantilever Actuation

Electrostatic Parallel-Plate Actuation

Electrostatic force between parallel electrodes:

$$F_e = \frac{1}{2} \frac{\epsilon AV^2}{(g_0 - x)^2}$$

Mechanical restoring force of cantilever:

$$F_m = kx$$

Pull-in instability occurs when:

$$\frac{dF_e}{dx} > \frac{dF_m}{dx}$$

For parallel plates, this leads to:

$$x_{\text{pull-in}} = \frac{g_0}{3}$$

Key Limitations

- Highly nonlinear **force-displacement** relationship
- Usable sensing range limited to **~33%** of initial gap

Motivation: Geometry-Based Cantilever Solution

Proposed Solution: Non-Parallel (Tapered) Electrodes

Gap varies spatially along the cantilever:

$$g(y) = g_0 + \alpha y - x$$

Total capacitance becomes:

$$C(x) = \int_0^L \frac{\epsilon w dy}{g_0 + \alpha y - x} = \frac{\epsilon w}{\alpha} \ln \left(\frac{g_0 + \alpha L - x}{g_0 - x} \right)$$

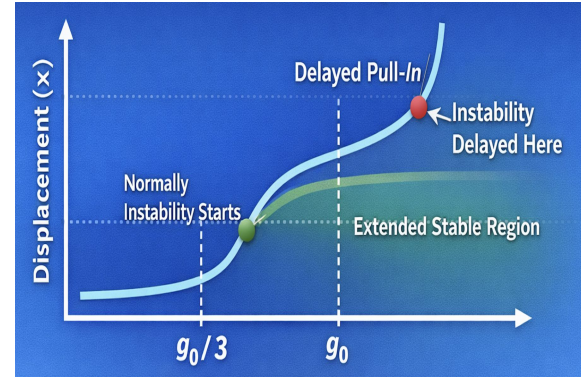
Electrostatic force:

$$F_e = -\frac{1}{2} V^2 \frac{dC}{dx}$$

$$F_e = \frac{1}{2} V^2 \frac{\epsilon w}{\alpha} \left(\frac{1}{g_0 - x} - \frac{1}{g_0 + \alpha L - x} \right)$$

Why this Is Better?

- Force gradient **increases** more slowly with displacement
- Pull-in instability is delayed beyond **$g_0/3$**
- Extended **quasi-linear** sensing range



Conclusion

- **MEMS cantilever-based pressure sensors** represent a major shift from bulky classical instruments to **miniaturized, high-precision sensing systems** with wide application potential.
- The **cantilever beam principle** efficiently converts tiny mechanical deformations into electrical signals, enabling **high sensitivity and fast response** for pressure and force measurement.
- Advances in **microfabrication, materials (silicon, silicon nitride), and capacitive readout electronics** have made MEMS sensors compact, reliable, and compatible with integrated circuits.
- **Miniaturization and cantilever arrays** significantly improve sensitivity and allow **multi-parameter and multiplexed sensing** on a single chip.
- To overcome limitations like pull-in instability and limited linear range, **geometry-based** designs such as **non-parallel (tapered) electrode gaps** are introduced, which **extend the sensing range, improve stability, and enhance performance** for next-generation MEMS sensors.